

Achyut Nair

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EDUCATION

University of Texas at Austin

Austin, TX

B.S. in Chemical Engineering, Minor in Economics; GPA: 3.82

Expected May 2028

- Affiliations: Longhorn Racing Combustion, AIChE Officer, Materials Science and Engineers, Energy Institute; Valero Summit Scholar and Dwight E. Huth Presidential Scholar

EXPERIENCE

Valero Energy Corporation

Ardmore, OK

Environmental Engineering Intern

May 2026 – Present

- Identified an outdated, manual Title V emissions calculation process and rebuilt it into modular, low-input Excel models, cutting reporting time by 50 hours and standardizing the workflow statewide
- Evaluated cost assumptions against operational requirements for the Influent Lift Station capital project and authored the technical and economic proposal that guided the investment decision
- Installed a Vapor Combustion Unit (VCU) flowmeter to enable direct process measurement, reducing compliance risk and improving emissions reporting accuracy
- Built real-time Critical Instrumentation monitoring software using Statistical Process Control (SPC) to flag flatline, drift, and out-of-control deviations

Longhorn Racing Combustion

Austin, TX

Composites Engineer

August 2025 – Present

- Built Python workflows to automate post-processing of ANSYS Fluent simulation outputs, cutting manual analysis time 85% and standardizing the team's pipeline
- Ran finite element analysis (FEA) on the undertray to design carbon fiber layups that met structural requirements within cost and material constraints
- Authored vacuum-assisted resin infusion documentation, standardizing fabrication steps across the composites team

Röchling Industrial Composites

Cleveland, OH

Research and Development Intern

May 2025 – August 2025

- Built ANSYS simulation workflows to supplement experimental testing, saving 105 hours and \$25,000 in Q3
- Analyzed experimental process data across 3 pultrusion lines to identify key performance parameters
- Performed FTIR and DSC characterization to evaluate resin chemistry and flow behavior

Energy Institute

Austin, TX

Student Assistant

August 2024 – Present

- Built a dedicated website for the Energy Week Conference, improving attendee experience and event communication
- Programmed a digital check-in system that cut system-wide delays by 2 hours daily
- Edited video content in Adobe Premiere that increased online visibility by 47%

PROJECTS

Longhorn Cube | Research Member

January 2025

- Reviewed literature on CO2 adsorbents and identified biochar as a cost-effective, scalable capture solution
- Modeled capture scenarios in Excel to minimize levelized cost of capture (LCOC) across sorbent efficiencies, plant scale, and throughput

TECHNICAL SKILLS

Programming & Computing: Python (NumPy, pandas, scikit-learn), Bash, Linux, MATLAB, HTML/CSS/JS

Process & Compliance: SPC, Six Sigma, Title V Emissions Compliance

Engineering & Analysis: ANSYS, DFT (VASP), SOLIDWORKS, Finite Element Analysis (FEA)

Materials & Characterization: FTIR, DSC, Carbon Fiber Composites, Fiberglass Composites

Digital & Content Tools: Adobe Premiere, HTML/CSS/JS, Web Development

AWARDS & INTERESTS

Achievements: National Speech and Debate Academic All American, 2X State Runner-up in Biotechnology Design, NASA Highschool Aerospace Scholars

Interests: Baking, Weight Lifting, Hiking, Podcasting (@Donesimply)